

LiteShip Pricing Engine

Technical Skills Evaluation Exercise

1. Introduction

The goal of this assignment is the development of the core logic of LiteShip: the rate calculation engine.

The challenge is to model a system where sales prices are not fixed, but derived from “Master” purchase price lists through configurable and granular calculation rules.

2. Functional Requirements: Price List Management

2.1 Hierarchical Modeling (Inheritance)

The system must implement a two-level structure:

Purchase Price List (Master):

Represents the contract signed with the courier (e.g., SDA). It defines the fundamental structure:

- weight ranges
- geographic zones
- base accessory costs

Sales Price List (Client-Specific):

Each client must be associated with a sales price list that inherits the structure from a Master purchase list.

Data duplication is not allowed; the sales price list must reference the Master list and apply markups.

2.2 Configurable Markup Engine

The candidate must implement sales price calculation logic based on three types of markup rules, definable at the Sales Price List level:

1. Base Freight Markup

A percentage applied to the base weight tariff (e.g., +15%).

2. Fixed Handling Fee

A fixed amount added to each shipment (e.g., + €1.50).

3. Granular Accessory Rules

Ability to define specific markup rules for accessory services, such as:

Fuel Surcharge: Variable percentage calculated on the base freight.

Cash on Delivery (COD): Threshold logic (e.g., €2.00 fixed up to €500, then 2% of the value).

2.3 Quote Calculation Logic

Given a shipment with a specific weight and accessory services, the system must:

- Identify the Master Purchase Price List and the related base cost
- Retrieve the markup rules from the client's Sales Price List
- Return a detailed quote showing separately: Base Cost, Markup, and Customer Total

3. Technical Specifications

Database Design:

Particular attention must be paid to the normalization of the Price List tables to support the Master/Sales reference.

Data Entry:

Create an interface or a set of APIs to define a new Sales Price List starting from an existing Master, setting the markup parameters.

4. Example Configuration

The candidate must implement dynamic calculation based on the following structure, where the sales price list inherits the structure from the master and applies the defined markup rules.

5. Expected Calculation Logic

For a shipment from client “Alfa” of 4 kg with Cash on Delivery (COD), the system should produce:

Master Base Cost:

€6.00 (Freight) + €0.60 (Fuel) + €1.50 (COD) = €8.10

Alfa Sales Price:

€7.20 (Freight +20%) + €0.72 (Fuel) + €2.00 (COD) + €1.00 (Handling Fee) = €10.92

Operating Margin:

€10.92 – €8.10 = €2.82

Cost Item	Purchase Price List (Master)	Client Price List 'Alfa' (Sales)
Freight 0–5 kg	€6.00	Master Cost + 20% Markup
Freight 5–10 kg	€9.00	Master Cost + 20% Markup
Handling Fee	€0.00	+ €1.00 (fixed fee)
Fuel Surcharge	10% (on freight)	Inherited from Master
Cash on Delivery (COD)	€1.50 fixed	Master Cost + €0.50

6. Evaluation

The following aspects will be evaluated:

- Cleanliness of the data model (avoid redundancy)
- Accuracy of mathematical calculations
- Ability of the system to handle dynamic weight ranges